Technical Report: Edge AI for Renewable Microgrids

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1. Executive Summary & Global Investment

Modern electrical grids are experiencing unprecedented decentralization due to rapid adoption of distributed photovoltaic and wind energy resources. In 2025, global investment in clean energy technologies reached a historic milestone of $450 billion. However, high-frequency voltage fluctuations require millisecond-level intelligent load balancing at the network edge.

This report outlines our solar-powered Edge AI architecture capable of autonomous islanding and real-time harmonic mitigation without centralized cloud roundtrips.

2. Deep Learning Models & Neural Quantization

To run modern transformer-based load forecasting models on resource-constrained embedded edge devices, we employed post-training quantization. Standard FP32 transformer weights were compressed to INT8 precision. Empirical benchmarks show that INT8 quantization reduced latency by 42% while retaining 99.1% accuracy relative to the uncompressed baseline model.

The neural inference engine utilizes a custom attention layer optimized for time-series telemetry streams, consuming an average of 4.8 watts of power during continuous evaluation.

3. Battery Management Systems & Cell Chemistry

Energy storage stability is paramount for microgrid longevity. Our study evaluated lithium iron phosphate (LiFePO4) cell chemistry under accelerated thermal stress cycles. In long-term cycle testing, LiFePO4 cells achieved 3,500 charge cycles before degrading to 80% capacity.

To prevent thermal runaway, an adaptive neural controller continuously monitors internal resistance and cell gradient temperatures, preemptively dampening charge rates when cell temperatures exceed 45 degrees Celsius.

4. Cyber-Physical Security & Uptime Resilience

Decentralized power nodes are vulnerable to distributed denial-of-service (DDoS) and man-in-the-middle attacks. Under our zero-trust initiative, named Project Aegis, all communications use cryptographic hardware secure elements. During adversarial red-team simulations, Project Aegis demonstrated 99.98% uptime against distributed simulated attacks.

Anomaly detection algorithms identify compromised nodes within 120 milliseconds, isolating affected subnets before false telemetry can cascade into blackouts.

5. Roadmap and Deployment Goals

The project consortium has established clear milestones for commercial scaling. Following successful pilot validations, the consortium plans to deploy 500 edge nodes across 12 countries by Q4 2026.

Future iterations will integrate bidirectional vehicle-to-grid (V2G) power routing and automated local carbon credit settlement using lightweight verifiable ledgers.